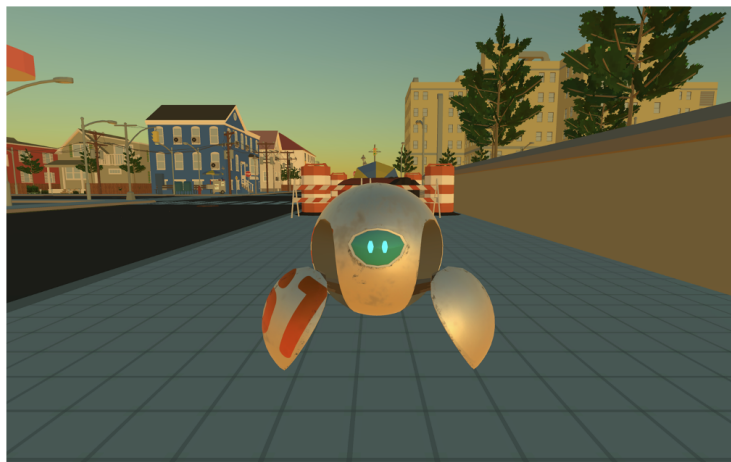


ParaTalk: A Real-Time Paralinguistic Dialogue System for Human-Agent Interaction

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(a) Display Screen



(b) Experiment View

Figure 1: (a) The display screen of a desktop application implementing the *ParaTalk* Dialogue System. We designed it for interactive communication with the robot in an outdoor environment. During the experiment, the screen display was fixed and unchanging to eliminate interaction elements other than audio. (b) A view of the preliminary experiment. Participants interacted face-to-face with a PC running a desktop application. The audio was played through earphones. The experiment was recorded simultaneously using an online meeting application, with images of the PC screen and the participant's frontal view.

ABSTRACT

Semantic-free utterances (SFUs) intuitively convey emotions and intentions, making them a promising tool for human-agent interaction. However, existing research has not addressed the challenge of generating context-aware SFU responses in real-time, limiting the adaptability and diversity of conversational systems. To address this gap, we propose *ParaTalk*. This real-time dialogue system dynamically generates six emotions (Happiness, Sadness, Fear, Anger, Surprise, Disgust) and five intents (Affirmation, Negation, Unsure, Question, Acknowledgement) based on user context. A preliminary user study demonstrated that *ParaTalk* effectively conveyed simple information through very short sounds (1-2 seconds), and maintained high user engagement. This suggests that SFU-based communication is feasible and demonstrates its potential effectiveness, particularly in scenarios where agents are required to exchange simple information.

Index Terms: Semantic-free utterances, Non-verbal communication, Human-agent interaction, Dialogue system

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1 INTRODUCTION

Advancements in automation and artificial intelligence have enabled humans to communicate with robots and Virtual Agents (VAs) through interactive dialogues [14]. These agents offer a range of benefits to users, including information retrieval [18], educational support [26], and psychological assistance [9]. In Virtual Reality (VR) and Mixed Reality (MR) environments, agents play a vital role in supporting learning and training [17, 24] and enhancing immersion [7], thereby facilitating human-computer interactions. By augmenting users' perceptual experiences, these agents have the potential to deliver cohesive cognitive and sensory experiences that seamlessly integrate virtual and physical spaces.

Voice interfaces utilizing natural language facilitate intuitive and efficient interactions with minimal training, simplifying communication with agents [33]. However, spoken language presents several challenges. While it effectively conveys complex information, it can also increase cognitive load [25], reduce usability in noisy environments [13], and create unrealistic expectations due to human-like voices [5, 11, 27]. These issues are particularly pronounced in VR/MR environments, where users must process complex visual and auditory information simultaneously, thereby limiting the practicality and usability of agents in such settings.

Semantic-Free Utterances (SFUs) have gained attention as a promising solution to these challenges. SFUs consist of sounds or vocalizations that lack linguistic meaning or dependence and are categorized into four types: Gibberish Speech (GS), Non-Linguistic Utterances (NLU), Musical Utterances (MU), and Paralinguistic

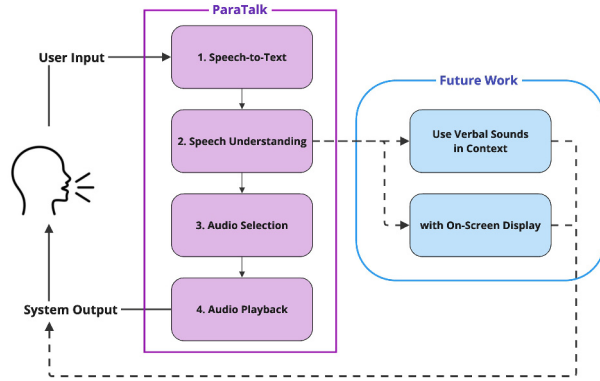


Figure 2: System flow of *ParaTalk*. After receiving user speech input, the processes enclosed in the purple box are executed, resulting in the playback of non-verbal sounds as the system output. The blue box indicates options for future work.

Utterances (PUs) [32]. SFUs have been shown to reduce cognitive load [3] and effectively convey emotions and intentions intuitively [21]. As such, they provide a flexible means for agents to express emotions and intentions, with their effectiveness well-documented [16, 22, 23]. In Mixed Reality (MR) environments, SFUs hold particular promise as auditory feedback mechanisms, with the potential to enhance multimodal experiences by integrating seamlessly with other sensory inputs [2, 29].

However, existing research has primarily focused on the impact of SFU-based agent utterances on users, with limited studies exploring real-time response generation using SFUs. For example, Kim et al. [10] demonstrated the context dependency of backchannel utterances associated with PUs (e.g., brief listener signals like “uh-huh” that convey attention or understanding). However, the study was limited to three Korean expressions (*ne*, *eo*, *eum*), which constrained the diversity of intention representation. Moreover, their study relied on the Wizard-of-Oz (WoZ) method, falling short of implementing an actual real-time dialogue system. Effective conversational communication is inherently interactive, requiring agents to comprehend the context of user utterances and generate responses that appropriately express emotions and intentions. Due to their inherent flexibility, SFUs enable diverse representations of emotions and intentions, offering the potential to go beyond the limitations of traditional backchannels and facilitate more dynamic and adaptive interactions.

This study proposes *ParaTalk*, a real-time dialogue system focusing on PUs, a type of SFUs. PUs consist of non-verbal sounds, such as exclamations and humming, that lack linguistic content and function as flexible backchannels in dialogue [16]. *ParaTalk* dynamically generates 30 types of responses by combining six emotions (Happiness, Sadness, Fear, Anger, Surprise, Disgust) with five intentions (Affirmation, Negation, Unsure, Question, Acknowledgement) based on the user’s conversational context. By integrating a Large Language Model (LLM), the system enables advanced, contextually adaptive response generation. This study aims to evaluate the effectiveness of an SFU-based dialogue system and explore its potential applications, contributing to the development of natural conversational interfaces with virtual agents.

2 DIALOGUE SYSTEM

The proposed dialogue system selects and plays appropriate non-verbal sounds based on the user’s verbal input, enabling responses aligned with the user’s utterances. This section provides a detailed explanation of the paralinguistic dialogue system.

2.1 System Flow

The system flow is depicted in Fig.2. The system processes user input and plays non-verbal sounds through the following four steps:

1. **Speech-to-Text:** The system converts user speech into text using Whisper [19], a speech recognition model developed by OpenAI.
2. **Speech Understanding:** The converted text is processed using the Realtime GPT model [20] via OpenAI’s API. This large language model (LLM) analyzes the input text using a prompt originally written in Japanese. When translated into English, the prompt is as follows:

You are an advanced conversational assistant. Based on the user’s utterance, select the appropriate emotion and intention. Emotions should be chosen from (Happiness, Sadness, Fear, Anger, Surprise, Disgust), and intentions from (Affirmation, Negation, Unsure, Question, Acknowledgement). Ensure the output includes only the emotion and intention, formatted as “Emotion, Intention”. Do not include any other information or format.

This prompt provides clear guidelines for classifying the emotion and intention associated with the user’s utterance. For the Preliminary User Study discussed later, the prompt was further refined to align with specific experimental scenarios or conversational contexts. Adjustments included narrowing the range of selectable emotions and intentions based on the scenario and controlling randomness to standardize experimental conditions.

3. **Audio Selection:** The system randomly selects audio clips corresponding to the chosen emotion and intention from a curated dataset. Each clip is pre-labeled with its associated emotion and intention.
4. **Audio Playback:** The system first plays the audio representing the emotion, followed by the audio conveying the intention. This approach ensures a clear distinction between the emotion and the specific intention.

In this study, we analyze user utterances using an LLM, which uniquely selects both emotions and intentions. However, certain user utterances, such as those requesting specific information or requiring visual content, may not be adequately addressed through non-verbal sounds alone. In such cases, the system could supplement non-verbal sounds with verbal outputs or screen displays to provide detailed information, as illustrated in Fig.2. Addressing these scenarios necessitates the selection of appropriate presentation methods based on the entropy or complexity conveyed in the user’s utterances. This challenge is beyond the scope of this study and is thus considered as future work.

Excluding the LLM component, the performance of this system heavily relies on the dataset of non-verbal sounds categorized by intentions and emotions, as well as their playback method. For playback, we adopted the simplest approach: sequentially playing the emotional sound followed by the intentional sound to effectively convey both aspects. Details of the dataset are described later.

2.2 Sounds Dataset

We developed a paralinguistic sound dataset that reflects users’ intentions and emotions, utilizing it as the primary audio resource for the system. To avoid monotone responses that might convey a cold or overly formal impression, we created a diverse range of sound variations [4, 31].

2.2.1 Data Creation

The classification of intentions was based on the study by Tran et al. [28], encompassing five categories: Affirmation, Negation, Unsure, Question, and Acknowledgement. The dataset was constructed with reference to [28] and [30], considering the types of PUs used in Japanese conversations. Eight audio samples were created for each intention class, recorded by two male and two female participants in their twenties. Examples of intentional sounds include Affirmation: *unun*, *soso*, Negation: *ie*, and Acknowledgement: *he*, *ne*, *ne*.

For emotion classification, Ekman's basic emotion model [6] was adopted, dividing emotions into six categories: Happiness, Sadness, Fear, Anger, Surprise, and Disgust. The emotional dataset utilized the online ASVP-ESD dataset [12] and the emotional corpus developed by Lima et al. [15], selecting eight audio samples per emotion. Gender balance was ensured during dataset construction. Most emotional sounds consisted of exclamations, with examples including Happiness: *yeah*, Fear: *hee*, and Surprise: *wow*.

Voice Conversion: The collected raw audio data underwent synthesis and transformation using a voice changer, followed by fine-tuning processes such as normalization and noise reduction. These steps aimed to homogenize the audio data and transform it into "robot-like" non-verbal sounds. As a result, 40 Intention Sounds across five categories and 48 Emotion Sounds across six categories were generated.

2.2.2 Data Selection Through Pilot Testing

Methodology: A pilot test was conducted to validate the dataset. Participants listened to randomly presented audio and classified them into one of the six emotion or five intention categories. Examples illustrating category characteristics were provided to ensure participants understood the task. A forced-choice method was employed. Participants included native Japanese speakers or foreigners proficient in Japanese.

Result: Eight participants ($M = 23.75, SD = 2.38, Min = 21, Max = 28, Male : Female = 5 : 3$) completed the pilot test. Recognition rates above 50% were deemed acceptable, and only sounds meeting this threshold were retained. Intention sounds were largely recognized accurately, and emotional sounds aligned well with their intended design. Ultimately, a minimum of four and a maximum of eight sounds were selected for each category, resulting in a final dataset of 33 Intention Sounds across five categories and 35 Emotion Sounds across six categories. Each selected sound had a duration of approximately one second, forming the final dataset used in the study.

3 PRELIMINARY USER STUDY

This study implemented a desktop application of the proposed dialogue system and conducted a preliminary interaction experiment to evaluate its effectiveness. The experiment was approved by the ethics review committee of the authors' affiliated institution.

3.1 Participants

Five participants whose native language is Japanese took part in the preliminary experiment ($M = 29.80, SD = 17.46, Min = 21, Max = 61, Male : Female = 4 : 1$). All participants were confirmed to have no significant auditory or visual impairments.

3.2 Procedure

The experimental procedure followed these steps. First, participants received an explanation of the experiment's purpose and procedures. They signed consent forms and completed a pre-experiment questionnaire, which included demographic information and questions about their experience with robots and VR systems. The experiment simulated interactions in an outdoor setting with a robot using a desktop application (see Fig.1-(a)). Participants sat in front of a laptop and wore earphones to standardize the auditory environment.

The interactions were recorded using an online meeting application to capture the system-participant interactions (see Fig.1-(b)). Participants engaged in the following two types of dialogues:

1. **Task-Oriented Dialogue:** Participants completed tasks where they selected the correct options for five topics based on non-verbal sounds presented by the robot. They were encouraged to freely ask questions to the robot and derive answers from the provided information.
2. **Non-Task-Oriented Dialogue:** Participants conducted a 10-turn conversation with the robot. During the first six turns, they followed specified content, while the remaining four turns allowed for free conversation. After each robot response, participants selected the perceived emotion and intention based on the non-verbal sounds provided.

After the experiment, participants completed a subjective evaluation questionnaire about the dialogue system. Additionally, a verbal interview provided an opportunity for participants to share overall impressions and feedback.

3.3 Evaluation

The collected data were analyzed using participants' questionnaire responses and dialogue logs with the robot. The evaluation focused on three aspects:

1. **Accuracy in Task-Oriented Dialogue:** Measured how accurately participants identified the correct options based on the robot's non-verbal sounds.
2. **Classification in Non-Task-Oriented Dialogue:** Assessed the accuracy of participants' emotion and intention selections based on the robot's responses.
3. **User Subjective Evaluation:** Participants rated the dialogue system using a questionnaire based on a Likert scale (1: Strongly Disagree - 7: Strongly Agree). The questionnaire comprised four System Features (System's Understanding, User's Understanding, Appropriateness, and Consistency), four Social Perception (Likability, Friendliness, Reliability, and Naturalness), and two Interaction Willingness (Willingness to Continue and Reuse). The questionnaire items referenced related studies [1, 8, 10, 16].

Through these evaluations, the study examined the effectiveness and areas for improvement of our dialogue system.

4 RESULT AND DISCUSSION

We obtained results for the three aspects set in the Evaluation section.

First, the accuracy rate in the task-oriented dialogue was remarkably high at 94% (Correct: 21, Incorrect: 1), excluding three system errors from 25 responses across five topics by five participants. This result demonstrates that even extremely brief utterances, such as non-verbal sounds, can effectively convey simple information.

In contrast, the performance of emotion and intention classification in the non-task-oriented dialogue resulted in a micro-F1 score of 0.56 for emotion and 0.49 for intention (see Fig.3). These scores are limited, with particularly low accuracy in classifying negative emotions. While this study does not primarily aim to achieve precise recognition of emotions and intentions, the possibility of significantly different meanings being misunderstood and transmitted highlights a limitation of the dataset. Although PUs inherently contain diverse semantic information, this study applied voice changers to standardize the audio. This process may have led to the loss of some information naturally present in human speech, potentially contributing to the difficulty in recognition.

Next, user subjective evaluations revealed relatively high scores: System Features ($M = 4.95, SD = 1.19$), Social Perception ($M = 4.95, SD = 1.50$), Interaction Willingness ($M = 5.4, SD = 1.51$), and

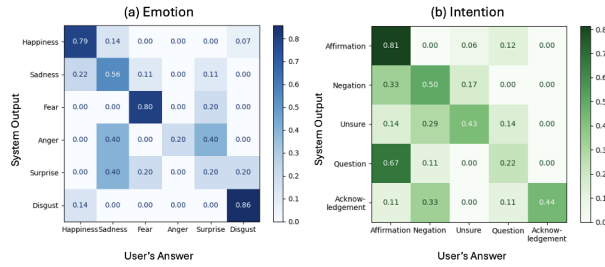


Figure 3: Normalized confusion matrices for Non-Task-Oriented Dialogue presented by the system output and user's answer. (a) Blue confusion matrix for emotions (Happiness, Sadness, Fear, Anger, Surprise, Disgust). (b) Green confusion matrix for intentions (Affirmation, Negation, Unsure, Question, Acknowledgement).

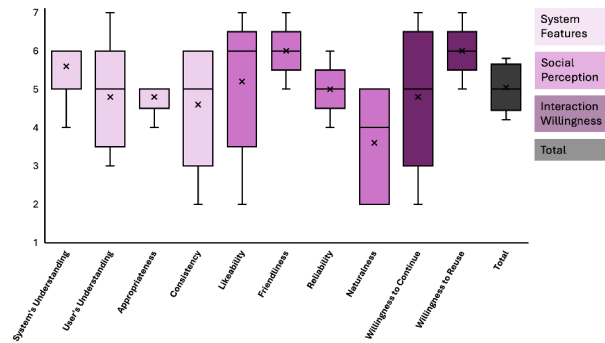


Figure 4: Box plot showing the User Subjective Evaluation Questionnaire results. Questions are color-coded by category: System Features, Social Perception, Interaction Willingness, and Total.

Overall ($M = 5.04, SD = 1.37$). Notably, "Interaction Willingness" received the highest score, indicating that users found interactions with the system comfortable. However, the large standard deviations suggest that some users were less satisfied. Additionally, the "Naturalness" item scored a low value of 3.6, highlighting a significant challenge. Participants commented that the current system's linear playback of emotional and intentional sounds felt unnatural. Considering that humans can simultaneously express emotions and intentions through non-verbal sounds, participants emphasized the need for the system to replicate this capability.

Based on these findings, future work should prioritize developing methods for generating natural non-verbal sounds that integrate emotions and intentions. Such advancements are expected to make the system's responses more intuitive and natural, enhancing the overall user experience.

5 CONCLUSION

This study proposed a paralinguistic dialogue system, *ParaTalk*, utilizing SFUs. The system dynamically generates real-time non-verbal sounds by analyzing user utterances and combining six emotions (Happiness, Sadness, Fear, Anger, Surprise, Disgust) with five intentions (Affirmation, Negation, Unsure, Question, Acknowledgement). By incorporating speech recognition technology and a LLM, the system achieves diverse and contextually adaptive responses, surpassing the limitations of conventional backchannels. A pilot test was conducted to select highly recognizable sounds, resulting in the construction of a dataset comprising 33 emotional sounds and 35 intentional sounds. In a preliminary user study, the system demonstrated a high accuracy rate of 94% in task-oriented dialogue,

confirming the effectiveness of non-verbal sounds for information transmission. However, challenges remain in the classification accuracy of emotions and intentions during non-task-oriented dialogue. Despite these limitations, subjective evaluations highlighted strong user engagement, with high ratings for dialogue willingness. The findings of this study indicate the potential for *ParaTalk* to serve as an innovative dialogue interface in applications such as navigation and monitoring systems. Future work will focus on developing methods for generating natural sounds that integrate emotions and intentions, as well as improving the diversity and appropriateness of system responses.

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